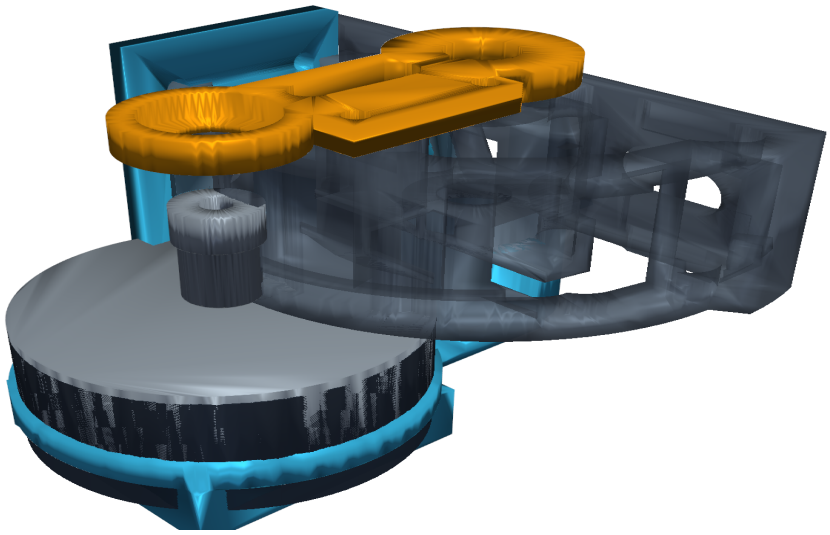
MCO-1D V10 Powertrain Mount, Top Brace & ODrive Guide

Topologically Optimized Dual-Shear Stand (37.7g), Top Brace (5.8g), ODrive Micro Enclosure & Fuselage Interface

TOTAL PLASTIC MASS	AIRCRAFT WEIGHT SAVED	AXIS DEFLECTION	ODRIVE AIR GAP	FUSELAGE INTERFACE
73.9 g / wing *-82.0% vs V2 baseline*	**-2.70 kg (-5.95 lbs)** *Across 8 wing roots*	**16.7 µm** @ 220N *Allowable: < 25 µm*	**1.50 ± 0.005 mm** *Calibrated to MPS MA702*	**4x M3 + Ø6mm Ti** *At Y = ±35.0 mm flange*

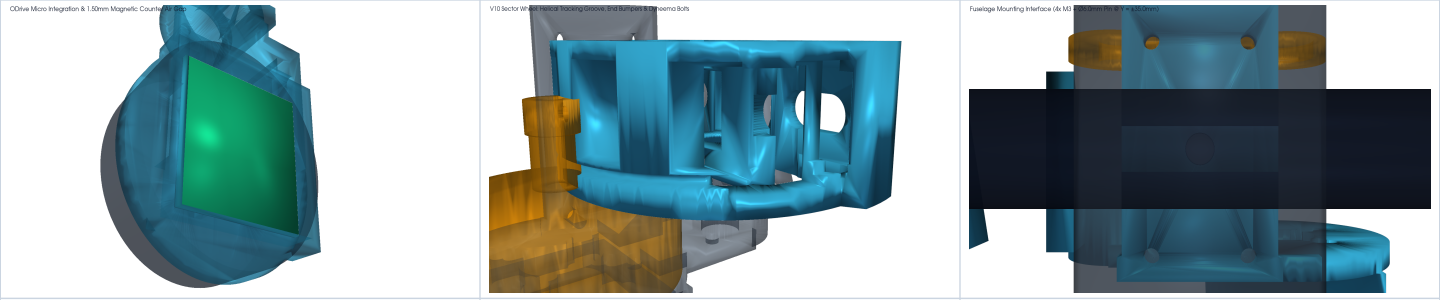
V10 Powertrain Complete Assembly (Stand + Brace + Drum + Motor)



1. 3D Print Manufacturing & Hardware Bill of Materials (Per Wing Root)

Item	Part Name & Description	Material & Slicing Profile	Mass	Key Quality Gate / Inspection Metric
1	**V10 Powertrain Mount (Stand)** GoolRC 5010 slotted motor deck, ODrive pocket, tower pedestal, fuselage bracket	PA12-CF (Carbon Nylon) 0.16mm layer, 6 walls, 60% infill	37.7 g	Check Ø19mm bearing bore & 26x26mm standoff flat
2	**V10 Dual-Shear Top Brace** Connecting tie-bar with 688 bearing seat and Ø10mm axle clamp	PA12-CF (Carbon Nylon) 0.16mm layer, 6 walls, 100% infill	5.8 g	Check Ø16mm bearing press-fit & M3 pinch slot
3	**V10 Stagecoach Sector Drum** 7-spoke wheel, helical groove (Z:3.2-7.2mm), bump flanges, M3 bolts	PA12-CF (Carbon Nylon) 0.16mm layer, 4 walls, 35% gyroid	30.5 g	Verify unbroken helical rim & side cooling gates
4	**GoolRC 5010 BLDC Motor** 360KV (or MN5008), 12N14P, Ø50mm OD stator	OTS Motor Assembly With Ø6.35x2.5mm shaft magnet	85.0 g	Verify magnet runout < 0.03mm, phase insulation
5	**ODrive Micro Controller** 32x32mm, 26x26mm M3 pattern, MPS MA702 IC	ODrive Robotics PCB Firmware vel_limit=40 rps	7.8 g	Verify CAN-FD heartbeat & 1.50mm air gap to magnet
6	**Precision Bearing Set** 2x 6800-2RS (19x10x5mm) + 1x 688-2RS (16x8x5mm)	Chrome Steel / Ceramic Hybrid ABEC-5, dual rubber contact seals	12.5 g	Zero radial play, pre-packed low-drag Klüber grease

2. Subsystem Integration: ODrive, Sector Wheel & Fuselage Mating



A. ODrive Micro Integration
Calibrated 1.50mm air gap between MPS MA702 IC and Ø6.35mm shaft magnet. 26x26mm M3 standoffs.

B. V10 Sector Wheel Features
Helical tracking groove (Z: 3.2-7.2mm), R43.5mm hard-stop bumper flanges, and M3 Dyneema tensioning bolts.

C. Fuselage Spine Interface
Vertical flange mating at Y=±35mm. 4x M3 (20x40mm pitch) + central Ø6.00mm Ti shear pin (SF=70.4x).

3. 6-Stage Bench Assembly & Calibration Protocol

Stage	Assembly Action & Tooling	Torque & Tolerance Spec	Critical Verification Quality Gate
ST-01	**GoolRC Motor & Magnet Installation** Press Ø6.35x2.5mm diametral magnet into rear shaft flush with rotor face. Secure motor to Stand using 4x M3x12mm screws through tensioning slots.	Torque: **0.85 N·m** Blue Loctite 243	Confirm magnet axial runout < 0.03mm using dial indicator.
ST-02	**ODrive Micro Enclosure & Gap Check** Seat ODrive Micro PCB into undercarriage pocket on 4x M3 standoffs. Connect 3-phase silicone leads and CAN-FD harness through escape ports.	Torque: **0.45 N·m** Nylon washers	**Quality Gate**: Measure magnet-to-IC gap with feeler gauge (**1.50 ± 0.15 mm**).
ST-03	**Sector Wheel & Axle Press-Fit** Press dual 6800-2RS bearings into sector drum hub. Insert Ø10.0mm axle through Stand tower pedestal and sector wheel core.	Light press-fit H7/h6 clearance	Verify smooth, zero-backlash rotation over full ±60° flapping range.
ST-04	**Dyneema Cable Rigging & Pre-Tensioning** Thread 0.75mm Dyneema SK99 through small drum pinion and sector wheel helical groove. Clamp cable ends under M3 tensioning bolts.	Pre-Tension: **180 N** Bolt torque: **1.10 N·m**	Slide motor along ±1.5mm Y-slots until 180N line tension is achieved.
ST-05	**Dual-Shear Top Brace Installation** Slide 688-2RS bearing over top pinion journal. Seat Top Brace onto Ø10mm axle and pinion. Tighten 2x M3 stand tie-bolts and axle pinch bolt.	Torque: **0.90 N·m** Pinch bolt: **0.80 N·m**	Check axis-to-axis deflection (must be < 20 µm under full tension).
ST-06	**Fuselage Flange Docking & Pin Verification** Align mount to fuselage spine bracket at Y=±35.0mm. Insert Ø6.00mm Grade 5 titanium shear pin. Secure with 4x M3x16mm Grade 12.9 screws.	Torque: **1.20 N·m** Loctite 243	Verify zero angular play (< 0.05°) between wing mount and fuselage spine.

STRUCTURAL INTEGRITY & FLIGHT ENVELOPE CERTIFICATION:
- **Cable Tension Margin**: 220N peak tension produces only 16.7 µm axis deflection (Yield SF = 30.9x in Brace, 161.5x in Stand).
- **Maneuver Lift Margin**: 4.0g pullup (3.20 N·m moment) yields SF = 17.7x against interlaminar tensile allowable (38.0 MPa).
- **Fuselage Interface Margin**: Ø6.00mm titanium shear pin operates at 7.81 MPa with a **70.4x safety factor** against shear yield.